GROUP 22

**FINAL PROJECT TITLE: Convenience Store Market Basket Pattern Explorer**
**Scenario:** A convenience store wants to analyze customer transaction records to discover products that are frequently purchased together. The system uses association-rule mining to identify common product combinations and purchasing patterns.

**Dataset:** 50 transactions, 15 products
**Primary algorithm:** Apriori
**Comparison method:** Frequent itemsets by support count
**Development tools:** KNIME, Python, Streamlit
**KNIME** — Analyzes the transactions visually through connected nodes.
**Python** — Recreates the same analysis using code.
**Streamlit** — Provides a simple website where users can upload transactions, adjust settings, and see the results.

**ROLES BELOW:**

| Member | Primary responsibility | Secondary responsibility |
|---|---|---|
| Sison | Python + Streamlit | Project leadership and integration |
| Sablan | KNIME + Algorithm Evaluation | Results comparison |
| Satimbre | Dataset + Documentation | Testing and KNIME assistance |

**Dataset:** Synthetic convenience-store purchases

**Transactions:** 50

**Transaction IDs:** T001–T050

**Unique products:** 15

**Typical items per basket:** 2–5

**DOCUMENTATION BELOW:**

| PRODUCTS | |
|---|---|
| Product no. # | Product name |
| 1 | Soft Drink |
| 2 | Bread |
| 3 | Chips |
| 4 | Instant Noodles |
| 5 | Coffee |
| 6 | Milk |
| 7 | Biscuits |
| 8 | Eggs |
| 9 | Corned Beef |
| 10 | Sardines |
| 11 | Bottled Water |
| 12 | Iced Tea |
| 13 | Chocolate |
| 14 | Peanut Butter |
| 15 | Hotdog |

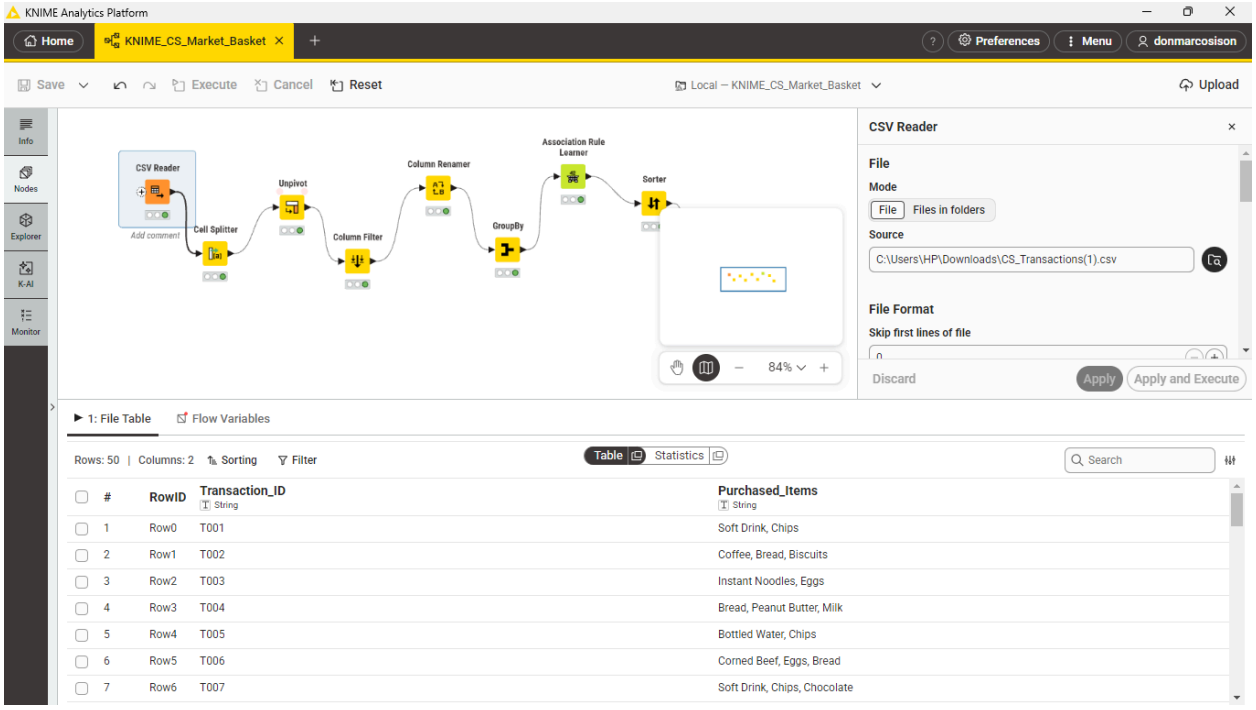
No item should appear twice in the same transaction. 2-5 different products. Transactions 1-50

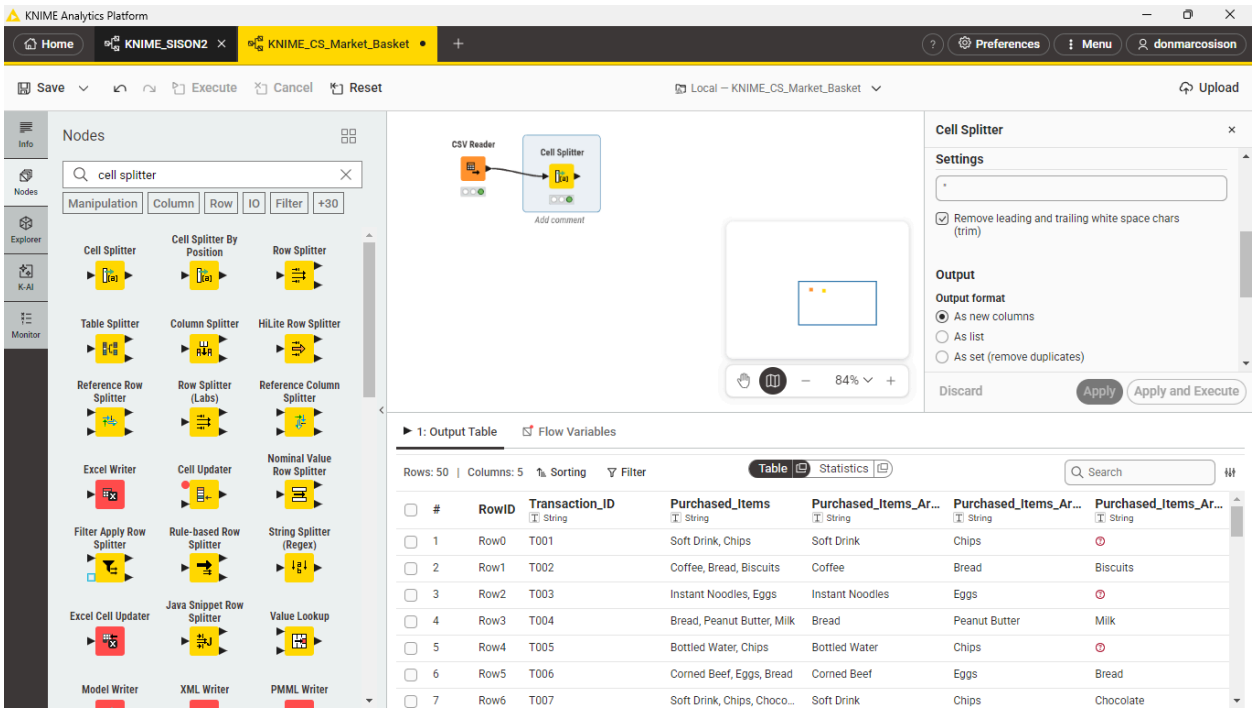
| CONVENIENCE STORE TRANSACTIONS | |
|---|---|
| **Transaction ID** | **Purchased Items** |
| T001 | Soft Drink, Chips |
| T002 | Coffee, Bread, Biscuits |
| T003 | Instant Noodles, Eggs |
| T004 | Bread, Peanut Butter, Milk |
| T005 | Bottled Water, Chips |
| T006 | Corned Beef, Eggs, Bread |

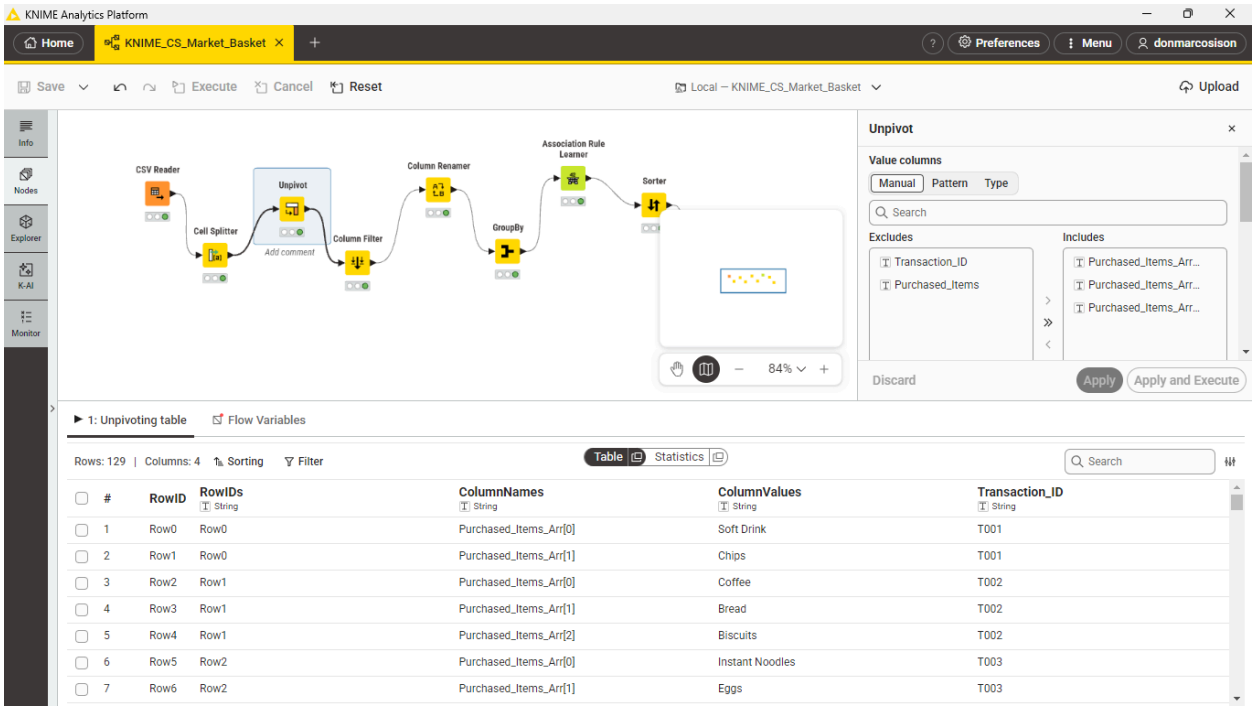
| | |
|---|---|
| T007 | Soft Drink, Chips, Chocolate |
| T008 | Coffee, Bread |
| T009 | Sardines, Eggs, Instant Noodles |
| T010 | Hotdog, Bread, Soft Drink |
| T011 | Coffee, Biscuits |
| T012 | Soft Drink, Chips, Biscuits |
| T013 | Bread, Peanut Butter |
| T014 | Instant Noodles, Eggs, Soft Drink |
| T015 | Bottled Water, Chocolate |
| T016 | Coffee, Bread, Milk |
| T017 | Corned Beef, Eggs |
| T018 | Soft Drink, Chips |
| T019 | Iced Tea, Biscuits, Chocolate |
| T020 | Hotdog, Bread |
| T021 | Instant Noodles, Eggs, Corned Beef |
| T022 | Coffee, Bread, Biscuits |
| T023 | Soft Drink, Chips, Chocolate |
| T024 | Sardines, Eggs |
| T025 | Bread, Peanut Butter, Milk |
| T026 | Bottled Water, Biscuits |
| T027 | Soft Drink, Chips, Bread |
| T028 | Coffee, Biscuits, Chocolate |
| T029 | Instant Noodles, Eggs |
| T030 | Hotdog, Bread, Soft Drink |
| T031 | Iced Tea, Chips |
| T032 | Corned Beef, Eggs, Bread |
| T033 | Coffee, Bread |

| | |
|---|---|
| T034 | Soft Drink, Chips, Chocolate |
| T035 | Sardines, Instant Noodles, Eggs |
| T036 | Bread, Peanut Butter |
| T037 | Bottled Water, Chips, Chocolate |
| T038 | Coffee, Bread, Biscuits |
| T039 | Instant Noodles, Eggs, Soft Drink |
| T040 | Hotdog, Bread |
| T041 | Soft Drink, Chips |
| T042 | Coffee, Biscuits |
| T043 | Bread, Peanut Butter, Milk |
| T044 | Corned Beef, Eggs |
| T045 | Iced Tea, Chips, Biscuits |
| T046 | Soft Drink, Chips, Chocolate |
| T047 | Sardines, Eggs, Bread |
| T048 | Coffee, Bread, Milk |
| T049 | Instant Noodles, Eggs |
| T050 | Hotdog, Bread, Soft Drink |

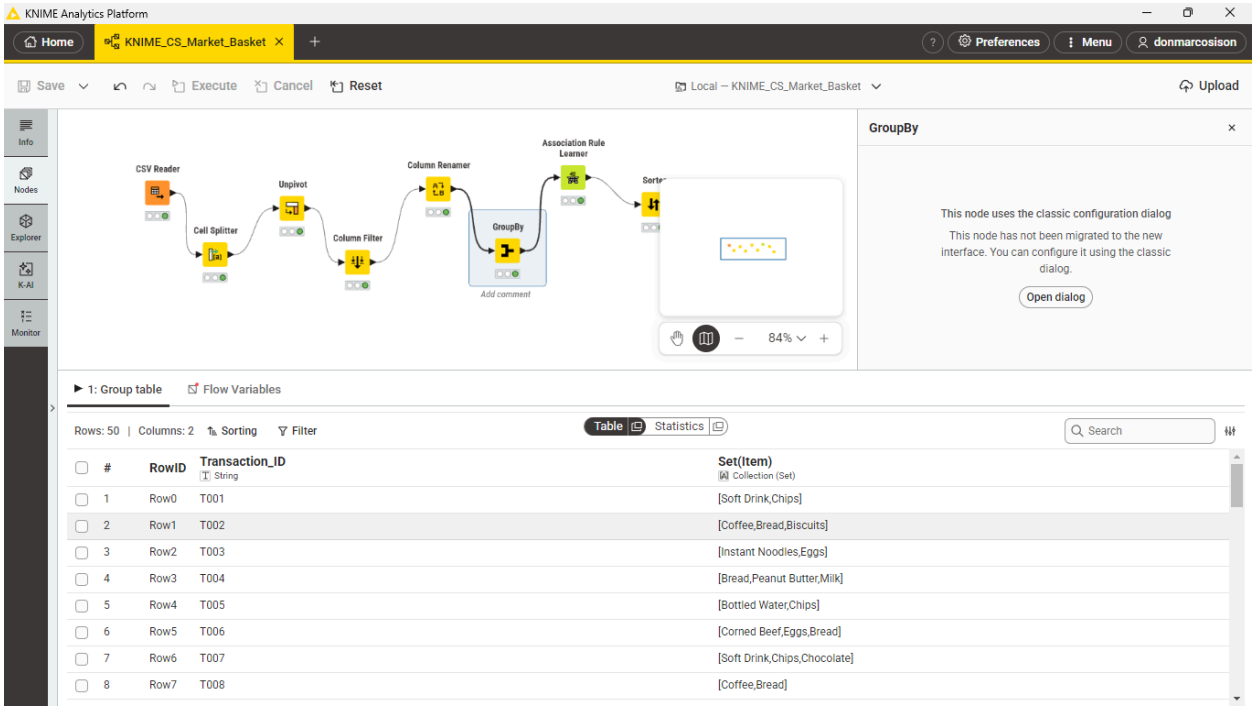
## KNIME:


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Files in folders
Source
C:\Users\HP\Downloads\CS_Transactions(1).csv
File Format
Skip first lines of file
Discard
Apply
Apply and Execute
1: File Table
Flow Variables
Rows: 50 | Columns: 2
Sorting
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# RowID Transaction_ID Purchased_Items
1 Row0 T001 Soft Drink, Chips
2 Row1 T002 Coffee, Bread, Biscuits
3 Row2 T003 Instant Noodles, Eggs
4 Row3 T004 Bread, Peanut Butter, Milk
5 Row4 T005 Bottled Water, Chips
6 Row5 T006 Corned Beef, Eggs, Bread
7 Row6 T007 Soft Drink, Chips, Chocolate

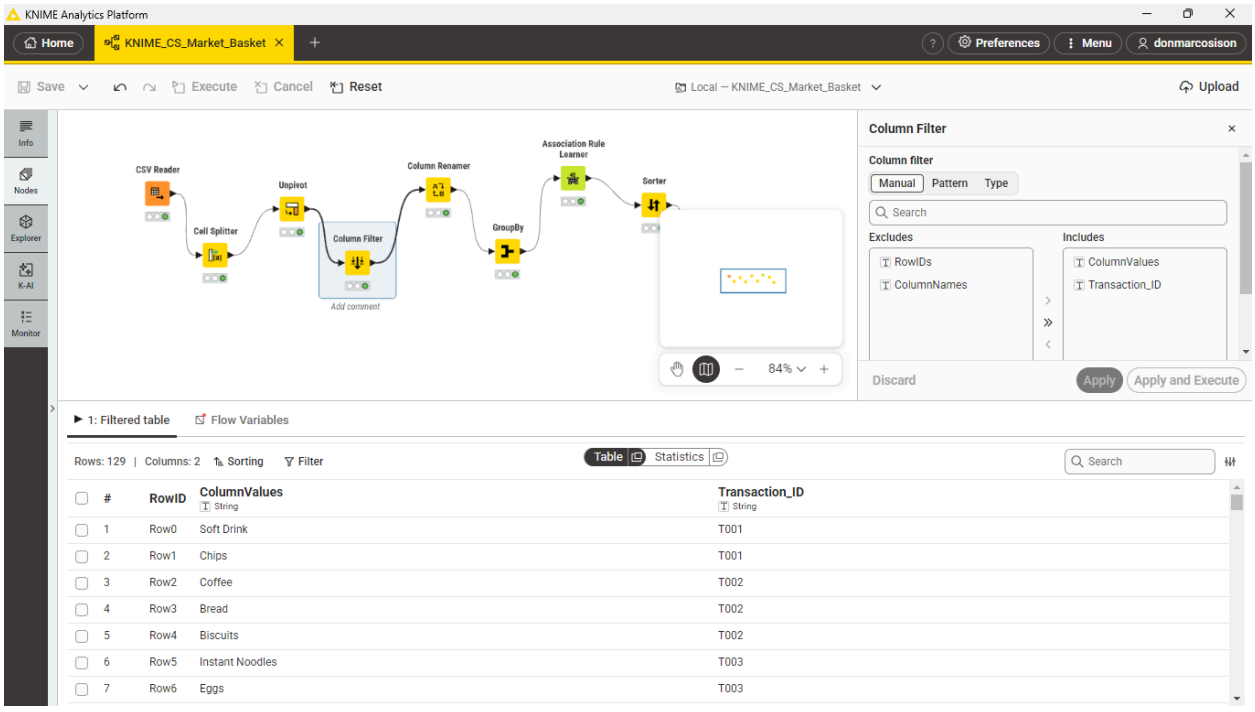

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Row Splitter
Table Splitter
Column Splitter
HiLite Row Splitter
Reference Row Splitter
Row Splitter (Labs)
Reference Column Splitter
Excel Writer
Cell Updater
Nominal Value Row Splitter
Filter Apply Row Splitter
Rule-based Row Splitter
String Splitter (Regex)
Excel Cell Updater
Java Snippet Row Splitter
Value Lookup
Model Writer
XML Writer
PMML Writer
CSV Reader
Cell Splitter
Add comment
Cell Splitter
Settings
Remove leading and trailing white space chars (trim)
Output
Output format
As new columns
As list
As set (remove duplicates)
Discard
Apply
Apply and Execute
1: Output Table
Flow Variables
Rows: 50 | Columns: 5
Sorting
Filter
Table
Statistics
Search
# RowID Transaction_ID Purchased_Items Purchased_Items_Ar... Purchased_Items_Ar... Purchased_Items_Ar...
1 Row0 T001 Soft Drink, Chips Soft Drink Chips
2 Row1 T002 Coffee, Bread, Biscuits Coffee Bread Biscuits
3 Row2 T003 Instant Noodles, Eggs Instant Noodles Eggs
4 Row3 T004 Bread, Peanut Butter, Milk Bread Peanut Butter Milk
5 Row4 T005 Bottled Water, Chips Bottled Water Chips
6 Row5 T006 Corned Beef, Eggs, Bread Corned Beef Eggs Bread
7 Row6 T007 Soft Drink, Chips, Choco... Soft Drink Chips Chocolate

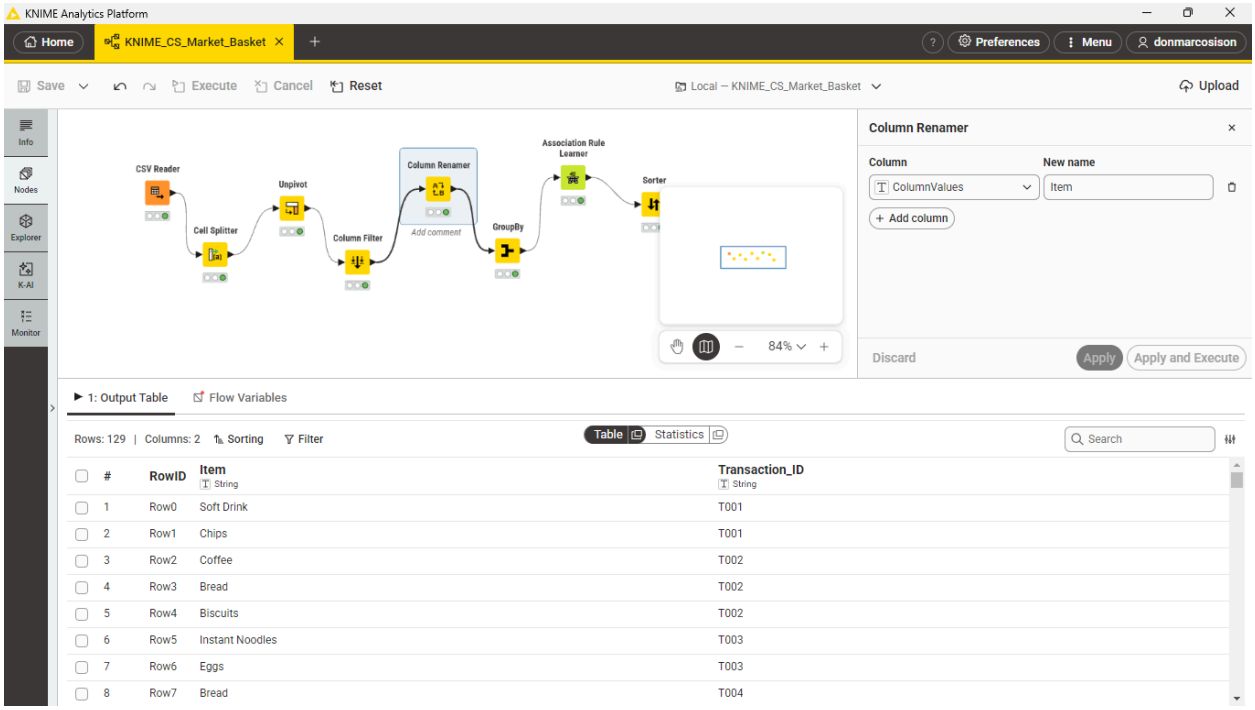


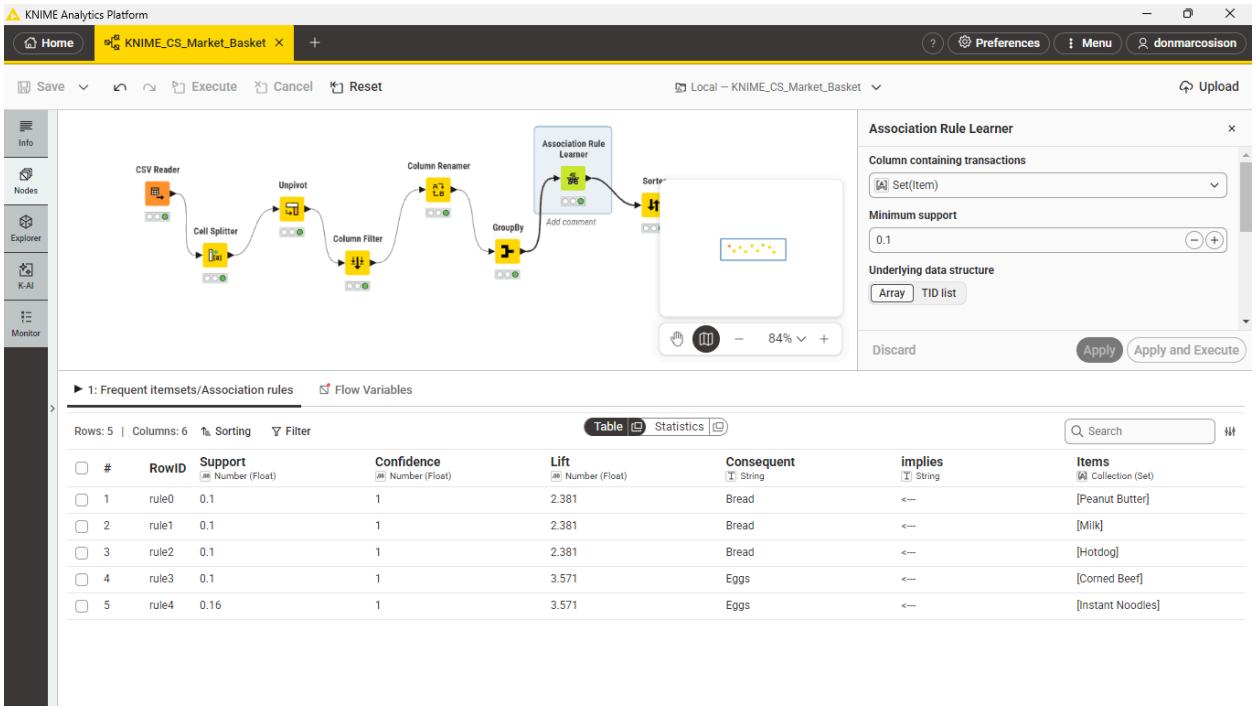

| # | RowID | RowIDs (String) | ColumnNames (String) | ColumnValues (String) | Transaction_ID (String) |
|---|---|---|---|---|---|
| 1 | Row0 | Row0 | Purchased_Items_Arr[0] | Soft Drink | T001 |
| 2 | Row1 | Row0 | Purchased_Items_Arr[1] | Chips | T001 |
| 3 | Row2 | Row1 | Purchased_Items_Arr[0] | Coffee | T002 |
| 4 | Row3 | Row1 | Purchased_Items_Arr[1] | Bread | T002 |
| 5 | Row4 | Row1 | Purchased_Items_Arr[2] | Biscuits | T002 |
| 6 | Row5 | Row2 | Purchased_Items_Arr[0] | Instant Noodles | T003 |
| 7 | Row6 | Row2 | Purchased_Items_Arr[1] | Eggs | T003 |

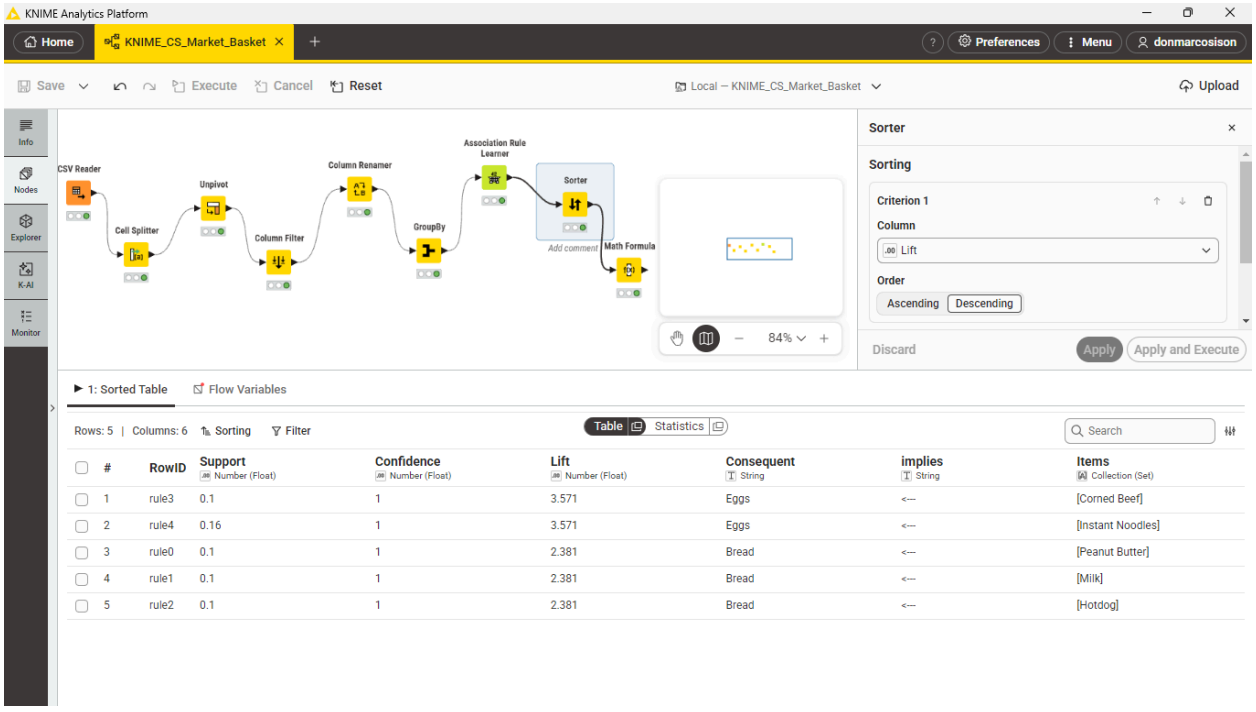


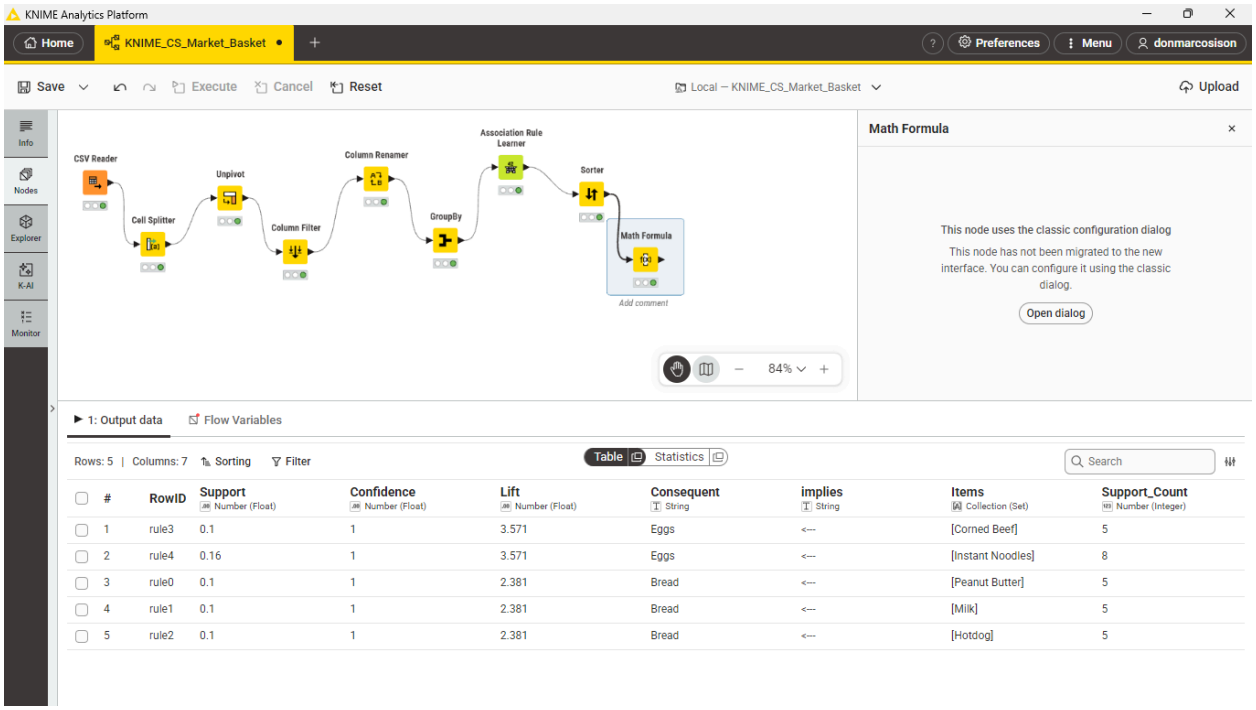

| # | RowID | Transaction_ID (String) | Set(Item) (Collection (Set)) |
|---|---|---|---|
| 1 | Row0 | T001 | [Soft Drink,Chips] |
| 2 | Row1 | T002 | [Coffee,Bread,Biscuits] |
| 3 | Row2 | T003 | [Instant Noodles,Eggs] |
| 4 | Row3 | T004 | [Bread,Peanut Butter,Milk] |
| 5 | Row4 | T005 | [Bottled Water,Chips] |
| 6 | Row5 | T006 | [Corned Beef,Eggs,Bread] |
| 7 | Row6 | T007 | [Soft Drink,Chips,Chocolate] |
| 8 | Row7 | T008 | [Coffee,Bread] |

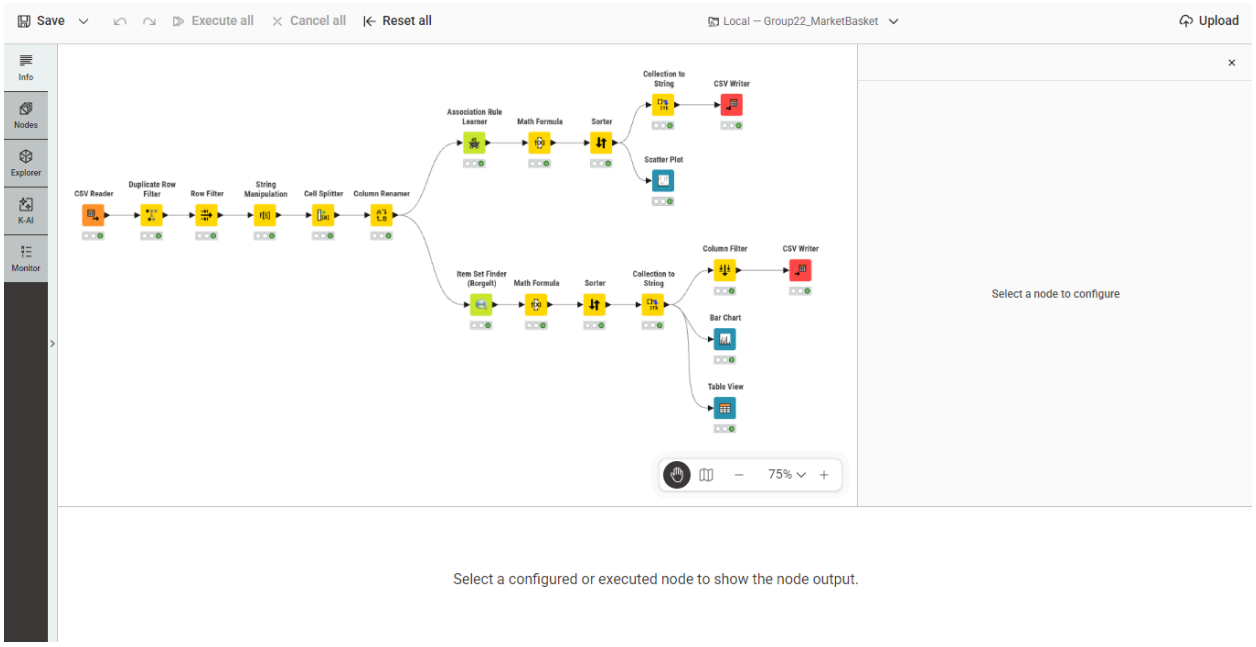

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Unpivot
Column Filter
Column Renamer
GroupBy
Association Rule Learner
Sorter
Add comment
84%
Column Filter
Column filter
Manual
Pattern
Type
Search
Excludes
RowIDs
ColumnNames
Includes
ColumnValues
Transaction_ID
Discard
Apply
Apply and Execute
1: Filtered table
Flow Variables
Rows: 129 | Columns: 2
Sorting
Filter
Table
Statistics
Search
# RowID ColumnValues String Transaction_ID String
1 Row0 Soft Drink T001
2 Row1 Chips T001
3 Row2 Coffee T002
4 Row3 Bread T002
5 Row4 Biscuits T002
6 Row5 Instant Noodles T003
7 Row6 Eggs T003


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Column Filter
Column Renamer
Add comment
GroupBy
Association Rule Learner
Sorter
84%
Column Renamer
Column
New name
ColumnValues
Item
Add column
Discard
Apply
Apply and Execute
1: Output Table
Flow Variables
Rows: 129 | Columns: 2
Sorting
Filter
Table
Statistics
Search
# RowID Item String Transaction_ID String
1 Row0 Soft Drink T001
2 Row1 Chips T001
3 Row2 Coffee T002
4 Row3 Bread T002
5 Row4 Biscuits T002
6 Row5 Instant Noodles T003
7 Row6 Eggs T003
8 Row7 Bread T004

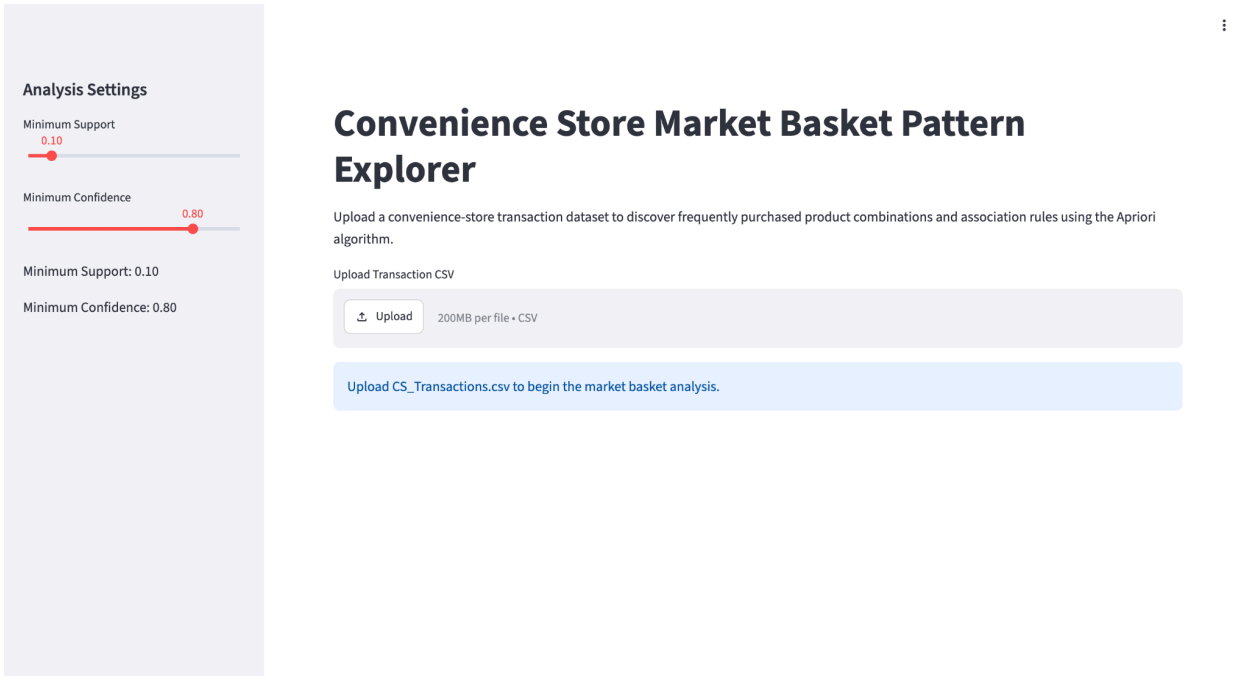

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Column Filter
Column Renamer
GroupBy
Association Rule Learner
Add comment
Association Rule Learner
Column containing transactions
Set(Item)
Minimum support
0.1
Underlying data structure
Array
TID list
Discard
Apply
Apply and Execute
1: Frequent itemsets/Association rules
Flow Variables
Rows: 5 | Columns: 6
Sorting
Filter
Table
Statistics
Search
# RowID Support Number (Float) Confidence Number (Float) Lift Number (Float) Consequent String implies String Items Collection (Set)
1 rule0 0.1 1 2.381 Bread <-- [Peanut Butter]
2 rule1 0.1 1 2.381 Bread <-- [Milk]
3 rule2 0.1 1 2.381 Bread <-- [Hotdog]
4 rule3 0.1 1 3.571 Eggs <-- [Corned Beef]
5 rule4 0.16 1 3.571 Eggs <-- [Instant Noodles]


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Association Rule Learner
Sorter
Add comment
Math Formula
Sorter
Sorting
Criterion 1
Column
Lift
Order
Ascending
Descending
Discard
Apply
Apply and Execute
1: Sorted Table
Flow Variables
Rows: 5 | Columns: 6
Sorting
Filter
Table
Statistics
Search
# RowID Support Number (Float) Confidence Number (Float) Lift Number (Float) Consequent String implies String Items Collection (Set)
1 rule3 0.1 1 3.571 Eggs <-- [Corned Beef]
2 rule4 0.16 1 3.571 Eggs <-- [Instant Noodles]
3 rule0 0.1 1 2.381 Bread <-- [Peanut Butter]
4 rule1 0.1 1 2.381 Bread <-- [Milk]
5 rule2 0.1 1 2.381 Bread <-- [Hotdog]

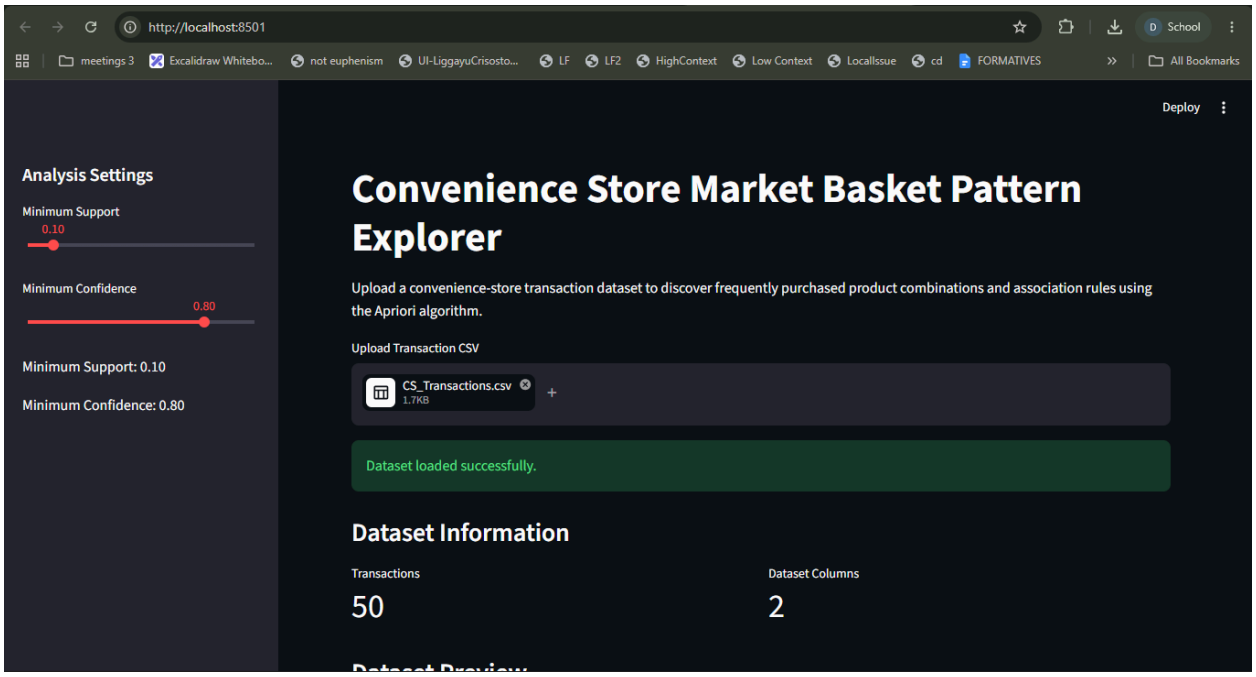

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GroupBy
Association Rule Learner
Sorter
Math Formula
Add comment
Math Formula
This node uses the classic configuration dialog
This node has not been migrated to the new interface. You can configure it using the classic dialog.
Open dialog
1: Output data
Flow Variables
Rows: 5 | Columns: 7
Sorting
Filter
Table
Statistics
Search
# | RowID | Support | Confidence | Lift | Consequent | implies | Items | Support_Count
1 | rule3 | 0.1 | 1 | 3.571 | Eggs | <--- | [Corned Beef] | 5
2 | rule4 | 0.16 | 1 | 3.571 | Eggs | <--- | [Instant Noodles] | 8
3 | rule0 | 0.1 | 1 | 2.381 | Bread | <--- | [Peanut Butter] | 5
4 | rule1 | 0.1 | 1 | 2.381 | Bread | <--- | [Milk] | 5
5 | rule2 | 0.1 | 1 | 2.381 | Bread | <--- | [Hotdog] | 5

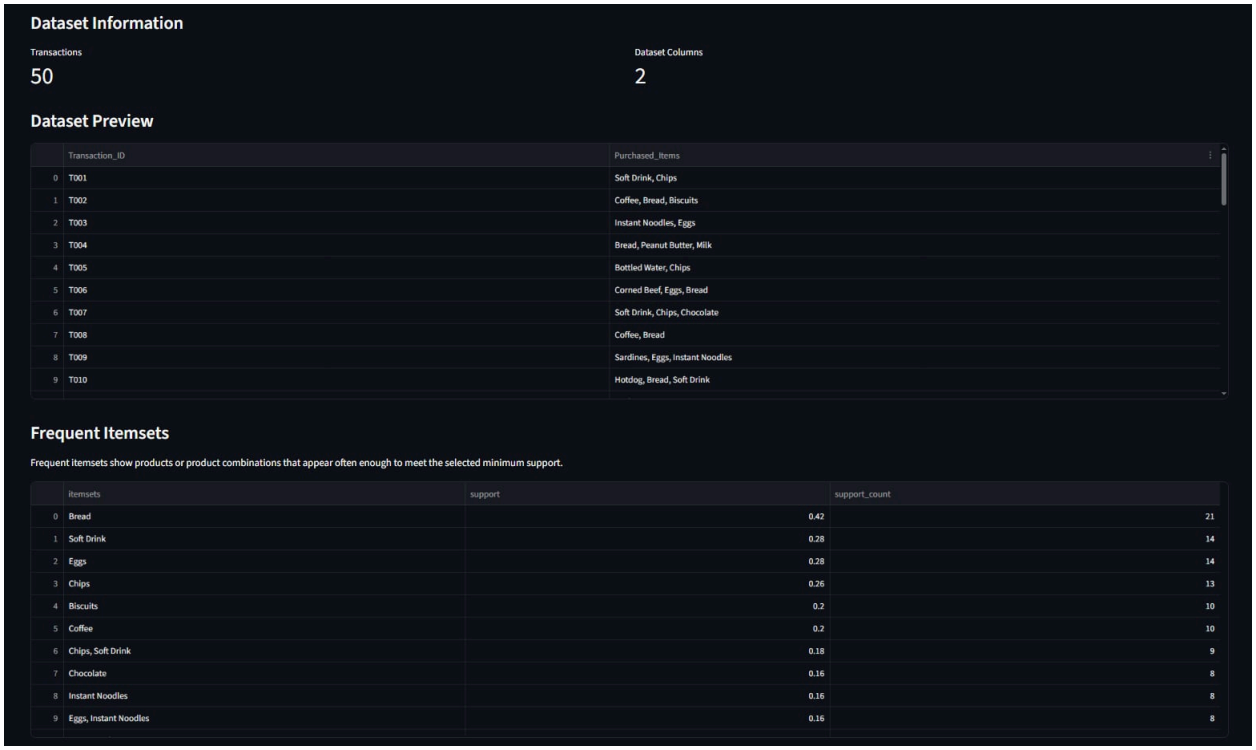

Save
Execute all
Cancel all
Reset all
Local – Group22_MarketBasket
Upload
CSV Reader
Duplicate Row Filter
Row Filter
String Manipulation
Cell Splitter
Column Renamer
Association Rule Learner
Math Formula
Sorter
Collection to String
CSV Writer
Scatter Plot
Item Set Finder (Borgelt)
Column Filter
Bar Chart
Table View
Select a node to configure
Select a configured or executed node to show the node output.

## STREAMLIT:




After uploading CS_Transactions.csv
Result below:





**Dataset Information**

Transactions
50

Dataset Columns
2

**Dataset Preview**

| | Transaction_ID | Purchased_Items |
|---|---|---|
| 0 | T001 | Soft Drink, Chips |
| 1 | T002 | Coffee, Bread, Biscuits |
| 2 | T003 | Instant Noodles, Eggs |
| 3 | T004 | Bread, Peanut Butter, Milk |
| 4 | T005 | Bottled Water, Chips |
| 5 | T006 | Corned Beef, Eggs, Bread |
| 6 | T007 | Soft Drink, Chips, Chocolate |
| 7 | T008 | Coffee, Bread |
| 8 | T009 | Sardines, Eggs, Instant Noodles |
| 9 | T010 | Hotdog, Bread, Soft Drink |

**Frequent Itemsets**

Frequent itemsets show products or product combinations that appear often enough to meet the selected minimum support.

| | itemsets | support | support_count |
|---|---|---|---|
| 0 | Bread | 0.42 | 21 |
| 1 | Soft Drink | 0.28 | 14 |
| 2 | Eggs | 0.28 | 14 |
| 3 | Chips | 0.26 | 13 |
| 4 | Biscuits | 0.2 | 10 |
| 5 | Coffee | 0.2 | 10 |
| 6 | Chips, Soft Drink | 0.18 | 9 |
| 7 | Chocolate | 0.16 | 8 |
| 8 | Instant Noodles | 0.16 | 8 |
| 9 | Eggs, Instant Noodles | 0.16 | 8 |



**Association Rules**

Association rules show relationships between products using support, confidence, and lift.

| | antecedents | consequents | support | confidence | lift | support_count |
|---|---|---|---|---|---|---|
| 0 | Instant Noodles | Eggs | 0.16 | 1 | 3.5714 | 8 |
| 1 | Corned Beef | Eggs | 0.1 | 1 | 3.5714 | 5 |
| 2 | Hotdog | Bread | 0.1 | 1 | 2.381 | 5 |
| 3 | Peanut Butter | Bread | 0.1 | 1 | 2.381 | 5 |
| 4 | Milk | Bread | 0.1 | 1 | 2.381 | 5 |

**This is our workflow below:**

**Phase 1 — Dataset preparation**

Create 50 synthetic transactions in a CSV file.

Assign IDs from T001 to T050.

Check missing values, spelling, and duplicate items.

Create a data dictionary explaining each field.

| Field Name | Data Type | Description | Example |
|---|---|---|---|
| **Transaction_ID** | String | Unique identifier assigned to each convenience-store transaction. | T001 |
| **Purchased_Items** | String/List | Products purchased together within a single transaction, separated by commas. | Soft Drink, Chips, Chocolate |

**Phase 2 — KNIME workflow**

Import the CSV using CSV Reader.

Convert purchased items into a transaction/item representation.

Generate frequent itemsets using the available association-mining nodes.

Configure minimum support and confidence.

Generate and sort association rules.

Record support counts for the comparison.

**Phase 3 — Python implementation**

Load and validate the same CSV using pandas.

Convert transactions into one-hot encoded item columns.

Run mlxtend Apriori.

Generate association rules.

Filter rules using support and confidence.

Calculate support count and lift, then export results.

**Phase 4 — Streamlit interface**

Add CSV upload and a default sample dataset.

Provide minimum support and confidence controls.

Display frequent itemsets and association rules.

Show support, confidence, lift, and support count.

Add downloadable result tables and friendly error messages.

**Phase 5 — Testing and comparison**

Test normal and missing-input cases.

Test duplicate items and invalid transaction data.

Test thresholds that produce no frequent itemsets or rules.

Compare KNIME and Python results with identical thresholds.

Write a short interpretation and limitations.

**Phase 6 — Documentation and presentation**

Compile dataset and data dictionary.

Capture KNIME and Streamlit screenshots.

Include the Python implementation and comparison table.

Explain key findings and limitations.

Rehearse each member's explanation.